

Software Requirements Specification (SRS)

Project: Vaultory — Small Business Inventory and Sales App (SBISA)

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Client / Sponsor	Small Business Retailer (Prof)
Base Document	BRD v3.4 (BRD-VAULTORY-001)

Revision History

Version	Date	Author	Description of Change
1.0	29/08/2026	Anoop (Solutions Architect)	Initial SRS derived from BRD v3.3
1.1	29/08/2026	Anoop (Solutions Architect)	Locked stack per BRD v3.4: React+Tailwind+shadcn/ui, Node.js backend + AI via Groq API, Supabase (Postgres/Auth incl. email OTP/Storage); resolved TBD-1 (Node) & TBD-2 (shadcn + Recharts)
1.2	12/09/2026	Anoop (Solutions Architect)	Synced with implemented behavior + DB schema v1.5 (<code>profiles</code> table, fractional quantities, PO receipts/returns, self-service signup, DB-first reports via views); added implementation-status summary (§1.4), updated auth/sales/PO/masking/indexes/TBD sections

Approvals

Role / Designation	Name	Signature	Date
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SIGN-OFF NOTICE: This SRS is the **technical baseline** for building Vaultory. It translates every BRD requirement into concrete, testable functional specifications. It is binding. **Any behavior not described here is Out of Scope**, exactly as in the BRD, and requires a Change Request.

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1. Introduction

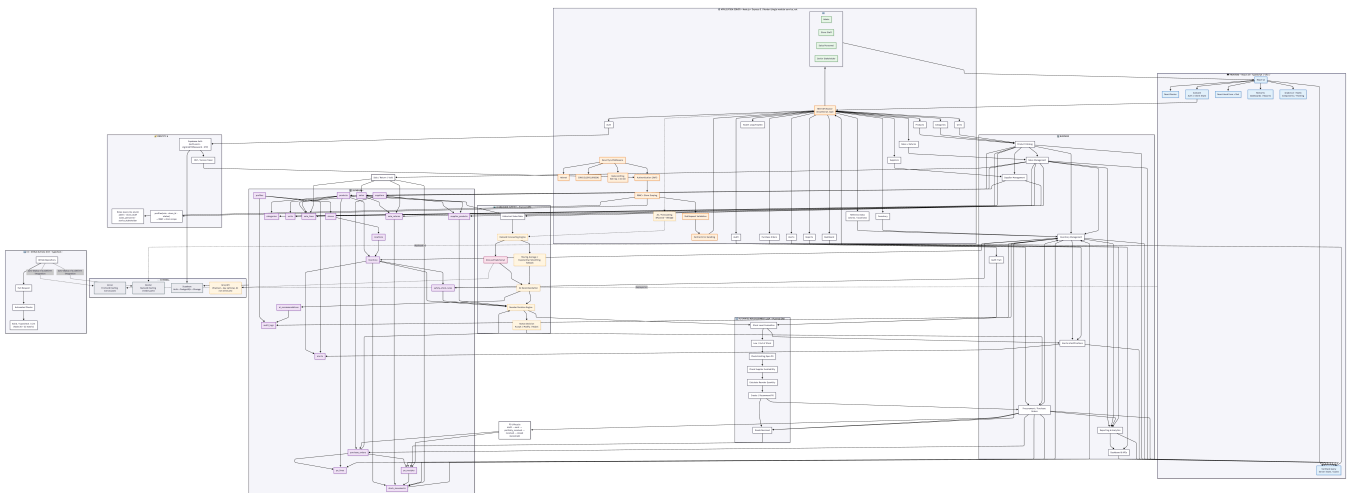


Figure 1 — Vaultory System Architecture — [view image on GitHub](#)

1.1 Purpose

This Software Requirements Specification (SRS) describes in technical detail **what the Vaultory application must do** so the development team (Tech Lead + team) can design, code, and test it, and the client can confirm the behavior. It is derived from, and fully consistent with, the **BRD v3.4**. Every requirement in the BRD is traced to a technical specification here (see Section 14).

1.2 Product Scope Summary

Vaultory is a web-based inventory and sales management application for a small retailer with **3 stores**. It provides:

- Multi-location inventory tracking (stores + warehouses).
- Sales recording with automatic stock deduction.
- Daily / quarterly / yearly sales reporting.
- Configurable safety stock & reorder points with alerts.

- **Automated AI ordering** when stock reaches the reorder point.
- **AI warehouse stock-level recommendations.**
- Store-wise sales performance views (sales personnel).
- Executive monitoring dashboards (senior stakeholders).
- Role-based access control (RBAC), data masking, audit logging.

1.3 Intended Audience

- **Client (Prof)** — to validate expected behavior.
- **Development team** — to implement and test.
- **QA / UAT** — to verify acceptance criteria.
- **PM / SM** — to estimate and plan (feeds the Sprint Planner).

1.4 Implementation Status (v1.2)

Convention: **Implemented** = behaviour is live in the current build; **Partial** = core path live, value-add pending; **Planned** = specified here but not yet delivered (tracked in §15).

Module	Status	Notes
Auth (email/password + JWT, OTP, forgot/reset)	Implemented	Public self-service signup <code>POST /api/auth/signup</code> always creates <code>store_staff</code> with a required store (server-side role clamp); Admin reassigns roles via the Users page. <code>GET /api/stores</code> is public to feed the signup dropdown.
Inventory (products/mask, categories, units, locations)	Implemented	Full-stack CRUD + tests; <code>cost_price</code> masking centralized in one utility (§7).
Stock (in/out/transfer/adjust, status view)	Implemented	Atomic <code>fn_mutate_stock()</code> ; immutable <code>stock_movements</code> audit; <code>compute_stock_status()</code> + <code>inventory_status</code> view.
Sales recording, returns, void	Implemented	Persistent via DB triggers; structured <code>sale_returns</code> / <code>sale_return_lines</code> .
Sales reports (day / qtr / year, store performance)	Implemented	DB-first: <code>daily_sales_summary</code> view (UTC); quarterly/yearly/store-performance aggregate queries.
Safety stock rules & alerts	Implemented	Rules + <code>alerts</code> module live; AI-triggered alert generation pending.
Suppliers & PO lifecycle	Implemented	PO numbering (<code>generate_po_number()</code>), atomic receipt (<code>fn_receive_po()</code>), per-delivery receipts history.
RBAC / data masking / store scoping	Implemented	Store-scoped dashboards, inventory and POs; masking via one shared utility.
AI ordering & warehouse recommendations (Groq)	Planned	Workflow specified in §8; Groq integration scheduled Sprint 3 (§15 TBD-7).
Value-adds: bulk import/export, onboarding wizard, email alerts	Partial / Planned	Schema-ready (<code>is_perishable</code> , <code>alerts</code> , audit); delivery depends on sprint capacity (§15).

2. Overall Description & Context

2.1 Product Perspective

Vaultory is an independent, self-contained web application. It does **not** integrate with external systems in this version (BRD §9). It consists of:

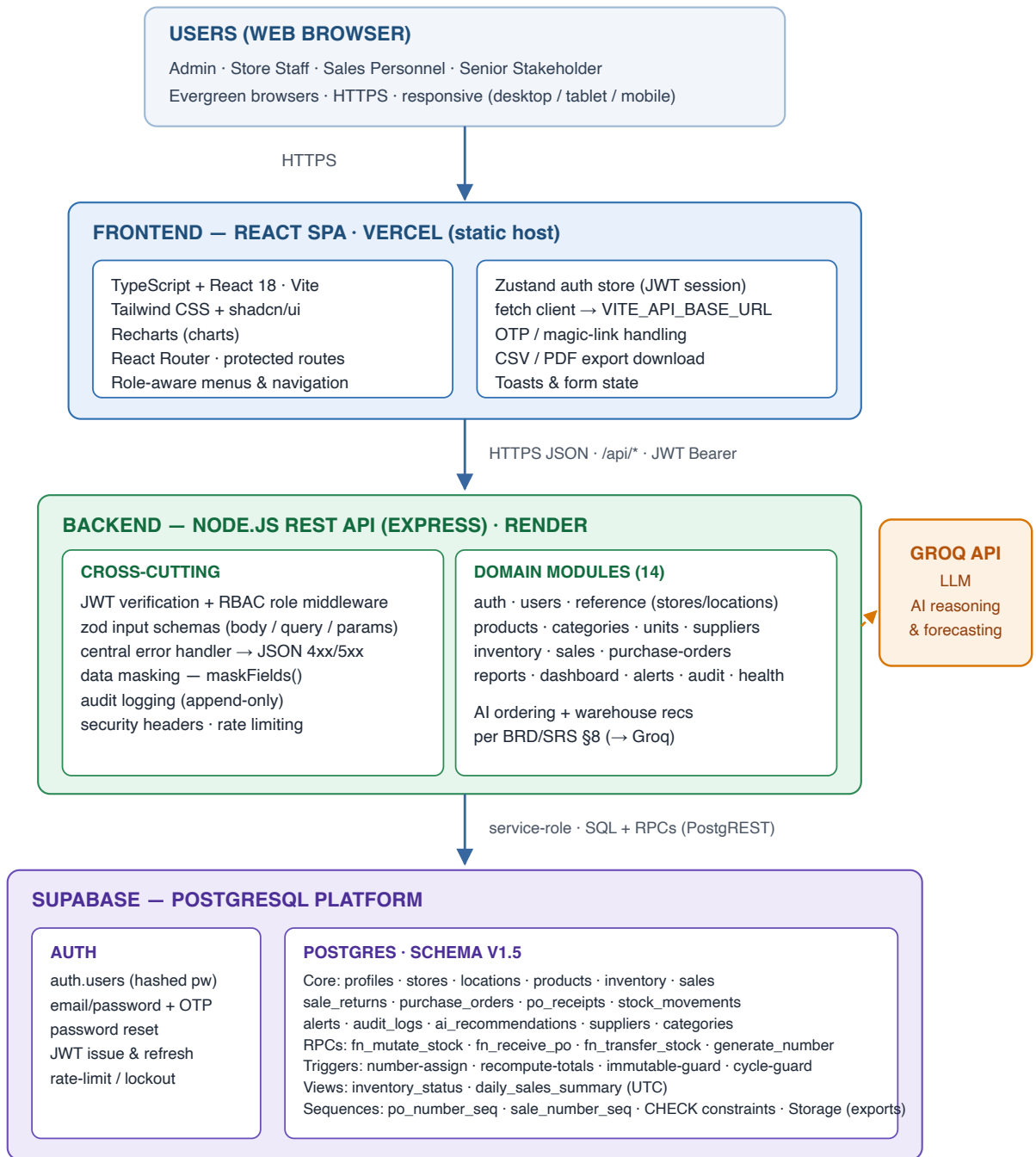


Figure 2 — Vaultory System Context (frontend ↔ API ↔ Supabase ↔ Groq)

2.2 User Classes & Characteristics

Class	Description	Skill Level
Admin	Configures products, users, suppliers, safety stock, rules	Basic–Intermediate
Store Staff	Records stock-in/out/transfer/adjustments, receives goods	Basic
Sales Personnel	Records sales, views store performance	Basic
Senior Stakeholder	Views dashboards & reports	Basic

2.3 Operating Environment

- **Browser:** modern evergreen browsers (Chrome 90+, Firefox 90+, Edge 90+, Safari 14+).
- **Devices:** responsive — desktop, tablet, mobile browser.
- **Network:** internet required; HTTPS only.
- **Hosting:** Vercel (frontend), Render (backend/AI), Supabase (Postgres/Auth/Storage), Groq API (AI inference) — free tier (BRD §8.5). GCP/AWS are Phase-2 post-handover only.

2.4 Design & Implementation Constraints

1. Frontend: **React (TypeScript) + Tailwind CSS + shadcn/ui**; charts via **Recharts**.
2. Backend: **Node.js** (Express **or** NestJS). (*Locked at BRD v3.4 — no Python.*)
3. Database/backend services: **Supabase** for **PostgreSQL + Auth** (email/password + email OTP) + **Object Storage**.
4. AI: Node.js AI service calling the **Groq API** (LLM reasoning/forecasting).
5. Deployment: **Vercel + Render + Supabase** free tier (BRD §8.5); Groq API key as a secret env var on Render.
6. All sensitive product information **masked**.
7. Single currency, single language.

2.5 Assumptions & Dependencies

Same as BRD §17 (3 stores, single currency/language, manual sales entry, client-provided master data, Vercel/Render/Supabase free-tier accounts, Groq free-tier API access).

3. Scope & References

3.1 In Scope

Identical to BRD §8 — all modules listed in the BRD (Inventory, Sales, Safety Stock, Procurement, AI Ordering, AI Warehouse Recommendations, Monitoring, RBAC, value-add modules).

3.2 Out of Scope

Identical to BRD §9 — L-1 through L-21 (no mobile apps, ecommerce, payments, POS, CRM, accounting, supplier portal, multi-currency, etc.).

3.3 References

- **BRD v3.4** — Business Requirements Document (BRD-VAULTORY-001).
- Problem Statement — Product 5: Small Business Inventory and Sales App (22/08/2026).

4. Functional Requirements — Detailed Specifications

This section expands each BRD functional requirement into **concrete technical behavior**: inputs, processing rules, outputs, validation, and error handling. Requirement IDs align with the BRD for traceability (`FR-<MODULE>-<seq>` = BRD ID; `.n` = technical sub-spec).

4.1 MODULE: Inventory Management

4.1.1 FR-INV-01 — Product (SKU) Master

- **Fields:** `sku_code` (unique CTEXT), `name`, `description`, `category_id` (FK, required), `unit_id` (FK, required), `cost_price` (masked, NUMERIC(14,2)), `sale_price` (NUMERIC(14,2)), `default_safety_stock` / `default_reorder_point` / `default_target_level` (NUMERIC(12,3) ≥ 0), `is_perishable` + `shelf_life_days`, `status` (active/archived), timestamps + `created_by`.
- **Validation rules:**
 - `sku_code` unique — reject duplicates with error.
 - `sale_price` ≥ 0, `cost_price` ≥ 0, `target` ≥ `reorder` ≥ `safety`.
 - Soft-deactivate (archive) only; never hard delete. Archived SKU blocked from new transactions; retained in history/reports.
- **Errors:** duplicate SKU → 409 SKU_ALREADY_EXISTS ; invalid price → 422 INVALID_PRICE .
- **Permissions:** create/edit → Admin only (BRD §12).

4.1.2 FR-INV-02 — Location Master (Stores & Warehouses)

- `locations` type ENUM: `store` | `warehouse`.
- Seed data: 3 stores (`Store A`, `Store B`, `Store C`) + 1 warehouse (`Central Warehouse`).
- Fields: `name`, `city`, `address`, `status` (active/inactive).
- Adding extra stores beyond 3 = Change Request (BRD L-14).

4.1.3 FR-INV-03 — Stock-on-Hand Tracking

- Table `inventory` : composite unique (`product_id`, `location_id`), `qty_on_hand` (integer, ≥0).
- Every transaction (stock-in/out/transfer/adjustment/sale) writes a `stock_movements` audit row and updates `qty_on_hand` accordingly.
- Stock view (read-only for authorized roles) returns: on-hand, `reorder_point`, `safety_stock`, `target_level`, status badge.
- **Status logic** — computed by `compute_stock_status(qty, reorder, target)` (exposed via the `inventory_status` view);
enum `stock_status` :
 - `out_of_stock` : `qty == 0`
 - `low` : `0 < qty ≤ reorder_point`
 - `in_stock` : `reorder_point < qty ≤ target_level` (any `qty` when `target = 0`)
 - `over_stock` : `qty > target_level` (`target > 0`)
- Level fallback: location-specific `safety_stock_rules` override product defaults (COALESCE in `inventory_status`).
- **All mutations go through `fn_mutate_stock()`** — one transaction that updates `inventory`, writes an immutable `stock_movements` row (with `qty_before` / `qty_after`), and refreshes timestamps. UPDATE/DELETE on `stock_movements` are blocked by trigger.

4.1.4 FR-INV-04 — Stock-In

- Input: `product_id`, `qty > 0`, `destination_location_id`, optional `po_id`, optional notes.
- Effect: `qty_on_hand += qty` at destination; if `po_id`, update PO received status (FR-PRO-05).
- Guard: destination active; `qty > 0` (fractional units supported).

4.1.5 FR-INV-05 — Stock-Out

- Input: `product_id`, `qty > 0`, `source_location_id`, `reason` (damage/loss/other), notes.
- Effect: `qty_on_hand -= qty`.
- Guard: **no negative stock** — if `qty > on-hand`, reject (409 INSUFFICIENT_STOCK) by default. (Config can downgrade to warn; default = block.)

4.1.6 FR-INV-06 — Stock Transfer

- Input: `product_id` , `qty` , `from_location_id` , `to_location_id` (must differ).
- Effect: `from -= qty` , `to += qty` in a single transaction (ACID); net zero.
- Guard: `qty ≤ from-location on-hand`.

4.1.7 FR-INV-07 — Stock Adjustment (Cycle Count)

- Input: `product_id` , `location_id` , `counted_qty` (≥ 0), `reason` (required).
- Behavior: compute `variance = counted_qty - qty_on_hand` . On confirm, set `qty_on_hand = counted_qty` , log adjustment with reason, actor, timestamp.
- Variance shown **before** confirmation.

4.1.8 FR-INV-08 — Status & Alerts

- After every stock mutation, re-evaluate status.
- On transition to `LOW` or `OUT` : create `alerts` row (`type = LOW_STOCK / OUT_OF_STOCK` , target roles = Admin + relevant store staff).
- Alerts marked `read / unread` ; list view in Alerts Center.

4.2 MODULE: Sales Management

4.2.1 FR-SAL-01 — Sale Recording

- Input: `store_id` , `sale_datetime` (server time default), optional `discount ≥ 0` , line items: [{`productId` , `qty`>0, `unitPrice`}] (client-sent totals ignored; server recomputes).
- Human-readable number: `sale_number` auto `SAL-<YYYY>--<seq>` from `generate_sale_number()` (sequence `sale_number_seq`); the `sales_assign_sale_number` trigger fills it when the caller omits it.
- Rules:
 - ≥ 1 line item required; `store_id` must reference a real store (trigger-checked).
 - `line_total = qty × unit_price` (generated column); `subtotal =  $\sum$  line_total` , `total = subtotal - discount` — maintained by recompute triggers; CHECK enforces `total ≥ 0` , `discount ≥ 0` , `total = subtotal - discount` .
 - **Stock deduction:** for each line, stock is decremented **atomically** at the sale's store via `fn_mutate_stock(type='sale', qty < 0)` (writes `qty_before / qty_after` + immutable movement).
 - If `qty > on-hand` → **block** by default (`409 INSUFFICIENT_STOCK`); identifies which product.
 - Sale is immutable after creation except void (FR-SAL-04); quantities support fractional units (`NUMERIC(12,3)`).
- Audit: each sale (`sale_created`) + stock movement recorded.

4.2.2 FR-SAL-02 — Sales Reports (Day / Quarter / Year)

- Endpoints/reports (all **DB-first** — computed by Postgres queries, not from in-memory state):
 - **Daily:** `daily_sales_summary` view — (`sale_datetime AT TIME ZONE 'UTC' )::DATE` grouping by store + product (timezone-safe; filters `store_id` , `product_id` , `date` applied over the view).
 - **Quarterly:** filter `store_id` , `product_id` , `quarter (YYYY-Qq)` — aggregate over `sales` + `sale_lines` .
 - **Yearly:** filter `store_id` , `product_id` , `year (YYYY)` — aggregate over `sales` + `sale_lines` .
 - **Store performance:** per-store aggregates (`sales` joined to `stores`), underlying `idx_sales_store_datetime` + `idx_sales_date_utc` indexes.
- Output metrics: `units_sold` , `sales_value` , grouped by store and product; sortable; export CSV/PDF.
- Voided sales are excluded (`WHERE status = 'active'`); returned qty reduces net via return records.

4.2.3 FR-SAL-03 — Store-wise Performance

- Sales personnel view: per-store dashboard (value, units, period); side-by-side store comparison; top products per store.

4.2.4 FR-SAL-04 — Void / Returns

- **Void:** only authorized users (Admin); require `reason`. Effect: sale marked `status=voided` with `voided_by` / `voided_at` / `void_reason` (CHECK-constrained) and excluded from report totals; stock restored **only for un-returned quantities** (already-returned lines are not re-credited, preventing over-stock); audit log `sale_voided`. Voiding a sale that already has a full line return is rejected (`RETURN_EXCEEDS_SOLD` logic) where applicable.
- **Return:** structured records `sale_returns` (header: `sale_id`, `store_id` must match the sale's store — trigger-checked, `reason`, `refund_amount` recomputed from lines by trigger) + `sale_return_lines` (each line references the original `sale_line_id`, `product_id` must match that line's product, total returned per sale line cannot exceed what was sold). Effect: stock restored via `fn_mutate_stock(type='sale_return', qty > 0)`; net sales totals reduce; `reason` captured. Returns against a **voided** sale are rejected (`SALE_IS_VOIDED`).
- **Persistence:** sale numbers, totals, returns, and void transitions are enforced **persistently via DB triggers/constraints** (not only in app code); `stock_movements` records carry `sale_id` / `sale_line_id` / `return_id` references.

4.3 MODULE: Safety Stock Management

4.3.1 FR-SST-01 — Configuration

- Per-product, per-location (or global default) values: `safety_stock`, `reorder_point`, `target_level`, `auto_order_enabled` (bool).
- Validation: `target_level ≥ reorder_point ≥ safety_stock ≥ 0`.
- Sources: manual entry **or** AI suggestion (accept → writes the suggested values).

4.3.2 FR-SST-02 — Tracking & Alerts

- Continuous comparison (after each mutation) of `qty_on_hand` vs `reorder_point`.
- `qty ≤ reorder_point` → product flagged + reorder evaluation triggered (Section 8.1) + alert created (target Admin; also relevant staff).

4.4 MODULE: Supplier & Procurement

4.4.1 FR-PRO-01 — Supplier Master

- Fields: `name`, `contact_person`, `phone`, `email`, `address`, `lead_time_days` (int>0), `status`, masked finance fields (e.g., credit terms).
- Mapping: `supplier_products(product_id, supplier_id)` many-to-many.

4.4.2 FR-PRO-02 / FR-PRO-03 — Purchase Order (Manual & Auto)

- PO fields: `po_number` (auto `PO-<YYYY>-<seq>` from `generate_po_number()` / `po_number_seq`), `supplier_id`, `destination_id`, `source` ENUM(manual, ai_auto), `status`, `order_date`, `expected_date = order_date + lead_time_days`, line items {`product_id`, `qty_ordered`, `qty_received`} (denormalized header totals `total_items` / `total_qty_ordered` / `total_qty_received` / `total_cost` maintained by `trigger_po_recompute_totals()`).
- **Auto-PO** generated by AI flow (Section 8.1); only one `is_preferred` supplier per product (partial unique index) drives deterministic selection.
- **Manual PO** created by authorized user with validated `qty > 0`; unique (`po_id`, `product_id`) line constraint.
- An open-PO partial index (`idx_po_open`) supports duplicate prevention (FR-PRO-06) and open-orders queries.

4.4.3 FR-PRO-04 — PO Lifecycle

- Status flow: `DRAFT` → `SENT` → `PARTIALLY_RECEIVED` → `RECEIVED` → `CLOSED` any state → `CANCELLED`.
- State transitions logged (actor, timestamp, old, new).

4.4.4 FR-PRO-05 — PO Receipt

- Goods-in is recorded as **per-delivery receipts**: each receipt creates a `po_receipts` header (batch/history) + `po_receipt_lines` (what actually arrived), increments `po_lines.qty_received` **by the delta**, and adds stock at the PO destination — all in one atomic transaction via `fn_receive_po()` (locks the PO line, validates product/PO/destination match, requires `earliest_expiry_date` for perishables).
- Over-receipt guard: CHECK `qty_received ≤ qty_ordered` on the line + composite FKs tie every receipt line to the same PO as both its receipt and its PO line (cross-PO integrity).
- If all lines fully received → `RECEIVED` ; partial → `PARTIALLY_RECEIVED` ; receipt history remains visible per delivery.

4.4.5 FR-PRO-06 — Duplicate Prevention

- Auto-PO suppressed if an open (non-closed/cancelled) PO for same `(product_id, destination_id)` already exists, unless config override.

4.5 MODULE: Value-Add Modules (BRD §8.1 / FR-CAT...FR-ONB)

4.5.1 FR-CAT — Categories & Units

- `categories` tree (self-referencing `parent_id`), `units` list. Assign product to one category + one unit.
- Reports group by category/unit. In-use values cannot be hard-deleted (soft archive).

4.5.2 FR-SUP — Supplier Performance (Value-add)

- Compute on-time delivery % per supplier over rolling window: `on_time = received_date ≤ expected_date` .
- Effective lead time for AI = rolling average of actual lead times (fall back to nominal if insufficient data).

4.5.3 FR-DSH — Dashboard Widgets

- KPI cards: total stock value, inventory turnover, low-stock count, out-of-stock count, today's sales.
- Role-constrained widget set; click-through drill-down.

4.5.4 FR-FSM — Fast/Slow Movers

- Classification by sales velocity (units sold per period) over rolling window (e.g., 90 days).
- Thresholds configurable (defaults: Fast $\geq$ threshold A, Slow $\leq$ threshold B).
- Slow movers flagged; fast movers prioritized in AI reorder/warehouse.

4.5.5 FR-EXP — Perishable Flag (advisory only)

- `is_perishable` bool + `shelf_life_days` (int). Warning/flag when estimated remaining shelf-life low. **No** batch/lot tracking (guardrail per BRD §10.13).
- Not in v1.0 core: marked `C` (Could) — delivered only if sprint capacity allows (see Sprint Planner).

4.5.6 FR-BULK — Import/Export

- Export: CSV of products, inventory, sales, POs (respect caller's masking permission).
- Import: products & opening stock via CSV template; server-side validation; per-row results (added/updated/failed with reasons); import session in audit log.
- Limits: max 5,000 rows per import; file $\leq$ 10 MB.

4.5.7 FR-ALR — Alert Preferences

- Per-user in-app preference toggles: low-stock, PO, expiry.
- Optional email alert to Admin/Seniors (config). No SMS/customer messaging.

4.5.8 FR-AUD — Audit Log Viewer

- Append-only log: actor, action, entity, entity_id, detail (JSON), created_at, ip (optional).
- Admin view with filters (actor, action, date range, entity).
- Masked sensitive values in logs.

4.5.9 FR-ONB — Onboarding Wizard

- Guided sequence: Location setup → Users → Products → Suppliers & mapping → Opening stock → Safety stock.
- Skippable, resumable, per-step validation with clear errors.

4.6 MODULE: User Management & RBAC

4.6.1 FR-USER-01 — Authentication

- Identity provider: **Supabase Auth (email/password + email OTP (magic link))**, accessed through backend endpoints: `POST /api/auth/signup`, `POST /api/auth/signin`, `POST /api/auth/otp (+ /verify-otp)`, `/forgot-password`, `/reset-password`, disabled in-app role selection — role is assigned server-side.
- **Self-service signup** (`POST /api/auth/signup`, **public**): always creates a `store_staff` profile with a **required** `storeId` (server-side clamp — a client-supplied role is ignored). Administrator reassigns roles/stores later via the Users page.
- `GET /api/stores` is **public** to feed the signup store dropdown (non-sensitive reference data); `/locations` and `/products` stay behind auth.
- Session: Supabase-issued **JWT (access token)** validated by the **Node backend** role middleware on every protected route; refresh token handled by Supabase.
- Logout invalidates the Supabase session; role-aware redirect after login; rate-limit/lockout configured in **Supabase Auth** settings.

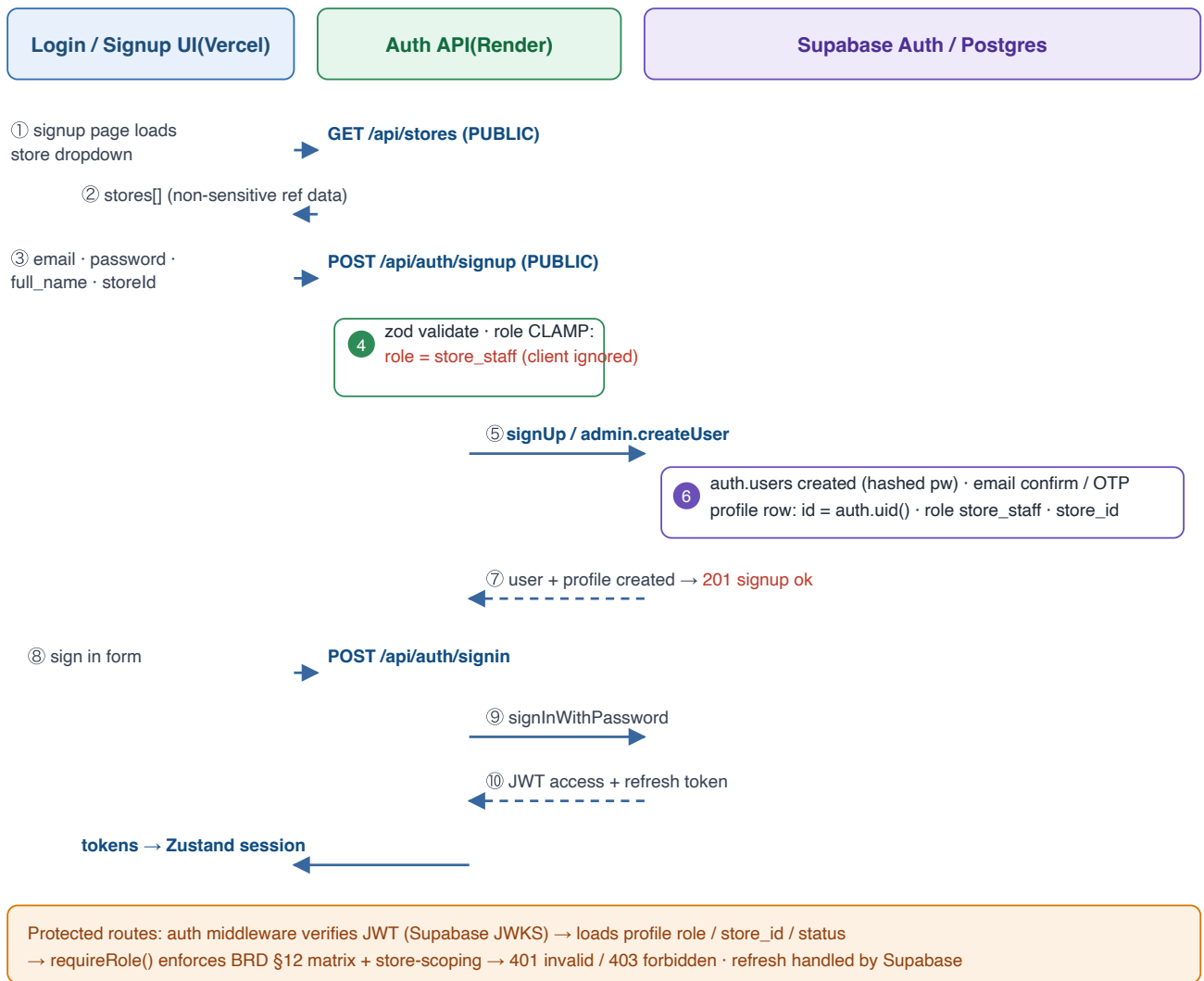


Figure 3 — Self-Service Signup & Authentication Flow

4.6.2 FR-USER-02 — Roles & Permissions

- Roles ENUM: ADMIN , STORE_STAFF , SALES_PERSONNEL , SENIOR_STAKEHOLDER (DB enum user_role ; profile defaults to store_staff).
- Protect **server-side** every route/method per BRD §12 matrix. Client hides menus; server enforces.
- Profile (profiles table, linked to auth.uid()) has store_id ; store-bound roles (Store Staff / Sales / Admin) are scoped to their store — dashboards, inventory, sales and POs are filtered by store.

4.6.3 FR-USER-03 — User Administration (Admin only)

- CRUD users (name, email, role, store, password reset), activate/deactivate.
- Cannot deactivate self or delete users (soft).

5. External Interface Requirements

5.1 User Interfaces

Web UI (responsive). Full per-screen specs in Section 11.

5.2 Hardware Interfaces

None (no POS, scanners, printers — BRD L-4).

5.3 Software Interfaces

- **REST API** between frontend and backend (JSON over HTTPS). Standards: RESTful resources, snake_case JSON, ISO-8601 timestamps, standardized error envelope `{ error: { code, message, details? } }`.
- **DB interface:** Node backend connects to **Supabase PostgreSQL** using the **Supabase JS/Postgres client** (or a Node driver/ORM such as Prisma) over the Supabase connection string.
- **AI module:** Node.js AI service that calls the **Groq API** (external LLM endpoints via HTTPS) for forecasting/recommendation reasoning, using secrets from env vars; deterministic rules stay in the Node backend.

5.4 Communication Interfaces

- HTTPS only (TLS 1.2+). CORS restricted to the Vercel frontend origin. Rate limiting on auth endpoints.

6. Data & Database Specifications

6.1 Schema (PostgreSQL via Supabase)

Auth note: user authentication/sessions live in **Supabase Auth** (`auth.users`), which also issues the OTP/magic-link and JWT tokens. The application `profiles` table below is the **profile/role table**, linked to the Supabase auth user (`id = auth.uid()` , FK with `ON DELETE RESTRICT`), storing `role` , `store_id` , and status for RBAC.

profiles	(id UUID PK = auth.users.id, email CTEXT, full_name, role user_role DEFAULT 'store_staff', store_id FK? → stores, gender, phone, status, last_login_at, created_at, updated_at) -- profile/role; auth (hashed password, sessions) in Supabase auth.users
stores	(id, code CTEXT UNIQUE, name, city, state, address, phone, email, status)
locations	(id, type ENUM(store,warehouse), store_id? FK → stores, code, name, is_default, status)
categories	(id, name, parent_id?, sort_order, status) -- tree; cycle-guard trigger
units	(id, name UNIQUE, abbreviation, status)
products	(id, sku_code CTEXT UNIQUE, name, description, category_id FK, unit_id FK, cost_price NUMERIC(14,2) MASKED, sale_price NUMERIC(14,2), default_safety_stock / default_reorder_point / default_target_level NUMERIC(12,3), is_perishable, shelf_life_days, status, created_by)
suppliers	(id, name, code, contact_person, phone, email, address, lead_time_days, payment_terms MASKED, credit_limit MASKED, total_pos, on_time_deliveries, avg_lead_time_days, status)
supplier_products	(supplier_id FK, product_id FK, unit_cost MASKED, lead_time_override, is_preferred, PRIMARY KEY(supplier_id, product_id)) -- ≤1 preferred per product
inventory	(product_id FK, location_id FK, qty_on_hand NUMERIC(12,3), earliest_expiry_date, last_counted_at, last_movement_at, PRIMARY KEY(product_id, location_id))
safety_stock_rules	(product_id FK, location_id FK?, safety_stock NUMERIC(12,3), reorder_point NUMERIC(12,3), target_level NUMERIC(12,3), auto_order_enabled, auto_approve)
purchase_orders	(id, po_number CTEXT UNIQUE, supplier_id FK, destination_id FK, source ENUM(manual,ai_auto), status po_status, order_date, expected_date, received_date, total_items, total_qty_ordered, total_qty_received, total_cost MASKED, created_by)
po_lines	(id, po_id FK, product_id FK, qty_ordered NUMERIC(12,3), qty_received NUMERIC(12,3), unit_cost MASKED, line_total GEN, notes) -- totals recompute by trigger
po_receipts	(id, po_id FK, received_by, received_at, notes) -- per-delivery receipts
po_receipt_lines	(id, receipt_id FK, po_line_id FK, po_id, product_id, qty_received NUMERIC(12,3)) -- composite FKs
sales	(id, sale_number CTEXT UNIQUE, store_id FK, sale_datetime, total_items, total_qty, subtotal, discount, total NUMERIC(14,2), status ENUM(active,voided), voided_by, voided_at, void_reason, created_by) -- totals recompute by trigger
sale_lines	(id, sale_id FK, product_id FK, qty NUMERIC(12,3), unit_price NUMERIC(14,2), line_total GEN)
sale_returns	(id, sale_id FK, store_id FK, return_datetime, reason, refund_amount, created_by)
sale_return_lines	(id, return_id FK, sale_line_id FK, product_id, qty_returned NUMERIC(12,3), unit_price, line_refund GEN) -- line must match sale_line product
stock_movements	(id, product_id FK, location_id FK, type movement_type, qty NUMERIC(12,3), qty_before, qty_after, sale_id?, sale_line_id?, po_id?, po_line_id?, return_id?, transfer_ref?, reason, notes, created_by, created_at) -- IMMUTABLE, via fn_mutate_stock()

```

alerts          (id, type alert_type, priority, title, message, product_id?, location_id?, po_id?,
                ai_recommendation_id?, target_roles user_role[], is_resolved, resolved_by, expires_at)
alert_reads     (alert_id FK, user_id FK, read_at, dismissed, PRIMARY KEY(alert_id, user_id))
alert_preferences(user_id PK, notify_low_stock, notify_out_of_stock, notify_po_*, notify_ai_recommendation,
                notify_expiry_warning, email_enabled, email_address)
ai_recommendations(id, type, status, product_id FK, location_id FK, recommended_value NUMERIC(12,3),
                current_value, reasoning, confidence, input_data JSONB, accepted_value,
                acted_on_by, acted_on_at, rejection_reason, resulting_po_id, expires_at)
audit_logs      (id, actor_id FK, actor_email, actor_role, action audit_action, entity, entity_id?,
                detail JSONB, ip_address, user_agent, created_at)          -- IMMUTABLE; sensitive values masked
app_settings    (key PK, value JSONB, description, updated_by, updated_at)
onboarding_progress(user_id PK, is_completed, current_step, step1..step6, skipped, completed_at)

```

Helper functions/views (source of truth: backend/src/db/schema.sql v1.5): sequences `po_number_seq` / `sale_number_seq` ; `generate_po_number()` , `generate_sale_number()` ; `compute_stock_status()` (pure); atomic mutators `fn_mutate_stock()` (inventory + movement, `SELECT FOR UPDATE`), `fn_receive_po()` (receipt + qty_received + stock), `fn_transfer_stock()` (two-location via shared `transfer_ref`); views `inventory_status` (COALESCE safety-level fallback, LATERAL rule pick) and `daily_sales_summary` (UTC date).

6.2 Indexes (as deployed)

- `inventory(product_id, location_id)` PK; `idx_inventory_location` for reverse lookups.
- `sales(store_id, sale_datetime)` (`idx_sales_store_datetime`) + expression index `idx_sales_date_utc` on `(sale_datetime AT TIME ZONE 'UTC')::DATE` for periodic reports.
- `sale_lines(sale_id)` , `sale_lines(product_id)` .
- `stock_movements(product_id, location_id, created_at)` + partial indexes on `sale_id` / `po_id` / `transfer_ref` .
- `po_lines(po_id)` , `po_lines(product_id)` ; `purchase_orders(supplier_id)` , `(destination_id)` , `(order_date)` , `(expected_date)` , partial `idx_po_open` (open POs only).
- `alerts(type)` , partial `(is_resolved)` WHERE `is_resolved = FALSE` , `(created_at)` , GIN `target_roles` .
- `profiles(role)` , partial `(store_id)` ; `products(name)` GIN trigram (fuzzy search); `categories(parent_id)` + partial unique top-level names.
- Uniqueness helpers: partial unique `uq_ssr_product_location` / `uq_ssr_product_global` (NULL-safe), `uq_locations_default_warehouse` , `uq_supplier_products_preferred` .
- `audit_logs(actor_id)` , `(action)` , `(entity, entity_id)` , `(created_at)` ; `ai_recommendations(product_id)` , `(type)` , `(status)` , `(created_at)` .

6.3 Data Integrity

- FK constraints on all references; soft deletes only (`status = 'archived'/'inactive'`) — no hard deletes of master data.
- Check constraints: `qty_on_hand ≥ 0` , movement signs per type, `total = subtotal - discount` , `target ≥ reorder ≥ safety` , `qty_received ≤ qty_ordered` , sale/PO state-transition timestamp rules.
- **Immutability:** `stock_movements` and `audit_logs` block `UPDATE` / `DELETE` via `trigger_immutable_guard` ; sale/PO totals recomputed by triggers on line changes.
- All stock mutations run through `fn_mutate_stock()` / `fn_receive_po()` / `fn_transfer_stock()` — single transaction, `SELECT FOR UPDATE` row locking prevents races.
- Sequences guarantee gapless-style `P0-<YYYY>--<seq>` / `SAL-<YYYY>--<seq>` numbering.

6.4 Backup

- Provider-managed backups where available on free tier + scheduled exports (nightly CSV/DB dump to object storage). Phase-2 targets: RPO ≤ 24h, RTO ≤ 4h.

7. Data Masking Specifications

7.1 Masked Fields (agreed at SRS)

- `products.cost_price`
- Computed `margin` (displayed where applicable)
- Supplier commercial/finance fields: `suppliers.payment_terms`, `suppliers.credit_limit`, `supplier_products.unit_cost`
- PO economics: `po_lines.unit_cost`, `purchase_orders.total_cost`
- Any field flagged sensitive in `app_settings['masked_fields']`

7.2 Masking Layers

1. **Database layer:** sensitive columns are stored with `- MASKED` comments and may be encrypted/tokenized (DBAs/read-only users cannot see raw values). RLS fields are prepared but RBAC+masking is enforced by the backend.
2. **Application/API layer — single centralized utility** (`backend/src/lib/masking.ts` → `maskFields(data, role, entity)`), called in **all response serializers**:
 - Authorized roles (ADMIN and explicitly authorized) receive clear-text **via a logged access path** (audit `sensitive_data_accessed`).
 - Unauthorized roles receive masked representation `"$*.*"` / `"....."` and masked value in exports too.
3. **Logs/exceptions:** masked values never logged; error messages and `audit_logs.detail` never contain raw sensitive values.

7.3 Masking Tests

- Test: forbidden role API call returns masked value; DB read of column returns masked/encrypted; export respects permission; audit log contains no raw values.

8. AI Engine Specifications

8.1 Automated Ordering Flow

Trigger (rerun after every stock-affecting event, plus a nightly sweep):

1. For each product where `qty_on_hand ≤ reorder_point` (at a location):
 - If `auto_order_enabled = false` → create LOW_STOCK alert only; stop.
 - If an open PO exists for (product, location) → skip (no duplicate) unless override.
 - If no supplier mapped for the product → create alert NO_SUPPLIER ; no PO.
2. Compute reorder quantity:

```
forecast = demand_forecast(product, horizon = effective_lead_time_days)
open_po = Σ open PO quantities (unreceived)
reorder_qty = max(0, target_level - qty_on_hand - open_po)
if forecast available AND auto_qty_mode='forecast':
    reorder_qty = max(reorder_qty, ceil(forecast))
```

3. Create PO (status DRAFT) to product's mapped supplier, `expected_date = today + lead_time`.
4. If `auto_approve = true` → SENT; else await Admin approval (low-stock alert with pending PO).

Edge cases:

- No sales history for product → fall back to `target_level - on_hand` (no forecast factor).
- Lead time unknown → use default 7 days + alert `MISSING_LEAD_TIME`.
- Multiple products, same supplier → batch into one PO where configured.

8.2 Warehouse Stock-Level Recommendation

For each (product, warehouse):

1. Forecast demand per unit time from historical sales (window default 90 days; min 14 days of history required; else “insufficient data” fallback).
2. `recommended_level = forecast × (lead_time + safety_buffer)` where `safety_buffer` = configurable (default 20%).
3. Clamp to `[safety_stock, target_level]` bounds; if unset, propose `2 × forecast` default.
4. Present with rationale string, e.g. *“Based on last 90 days, expected ~120 units/week. Recommend 240 units to cover 1.5-week lead time + 20% buffer.”*
5. User action: ACCEPT (writes to `safety_stock_rules` / inventory target), MODIFY (edits then accept), REJECT (logged).

8.3 Forecasting Method (v1.0)

- The AI service is a **Node.js module that calls the Groq API** (LLM inference) for demand forecasting and recommendation reasoning.
- Deterministic inputs (recent sales, window = 90 days default, min 14 days required) are passed to Groq; fallback to **simple moving average / exponential smoothing** in the Node backend if no LLM output or insufficient data.
- Trend/seasonality = optional refinement if data volume supports it (scope limited to auto-ordering + warehouse recommendation — BRD L-21).
- All AI inputs, computations (deterministic + Groq output), and outcomes recorded in `ai_recommendations` for audit.
- **Groq availability/resilience:** the AI reasoner is advisory; if the Groq API is unavailable (or free-tier rate limits are hit), the system falls back to deterministic forecasts and still functions (alerts/auto-order still work via rules).

8.4 AI Constraints

- AI is **advisory**; never executes an order without the auto-order consent flag.
- AI never bypasses RBAC or masking.
- Every recommendation explainable and auditable.
- **Groq API key** is stored as a secret environment variable (Render); never logged or committed.
- AI must **degrade gracefully** (deterministic fallback) if the Groq API is unavailable or rate-limited — core inventory/sales/alerts must not depend on the LLM.

9. Non-Functional Specifications

ID	Category	Specification (measurable)
NFR-1	Performance	Dashboard/inventory views ≤ 3s p95 under normal load.
NFR-2	Performance	Data reflected in reports ≤ 5 min (near real-time).
NFR-3	Scalability	3 stores, multiple warehouses, ≥ 50 concurrent users.
NFR-4	Availability	Phase-1 free tier: best effort (sleep/cold-start accepted). Phase-2: 99.5%.
NFR-5	Security	TLS 1.2+; encryption at rest.
NFR-6	Security	Authentication & OTP handled by Supabase Auth ; Node backend validates Supabase JWTs for RBAC; rate-limit/lockout configured in Supabase Auth.
NFR-7	Data Protection	Masking per Section 7.
NFR-8	Usability	Responsive UI; ≤3 clicks to core actions; clear labels.
NFR-9	Maintainability	Modular code, documented, linted; CI on push.

ID	Category	Specification (measurable)
NFR-10	Reliability	No data corruption on partial failure; DB transactions; error envelope.
NFR-11	Auditability	Append-only audit log for significant actions.
NFR-12	Backup	Provider backups (free tier) + scheduled exports; Phase-2 RPO≤24h RTO≤4h.
NFR-13	Compliance	Data-protection best practices on chosen providers.
NFR-14	Accessibility	Clear UI; single language (English).

10. Security Requirements

1. HTTPS only; HSTS enabled.
2. Authentication & password/OTP handling via **Supabase Auth** (no plaintext; passwords managed by Supabase).
3. **JWT issued by Supabase Auth**, validated by Node backend (RBAC + role middleware) on every protected route.
4. Rate limiting & lockout on login (configured in **Supabase Auth**); global API rate limit per IP on the Node backend.
5. Input validation on all inputs (server-side), parameterized SQL via the Supabase/Postgres client.
6. Masking of sensitive fields (Section 7).
7. CORS restricted to frontend origin; security headers (CSP, X-Frame-Options).
8. Dependency scanning in CI; `.env` secrets never committed — **Supabase** and **Groq** keys stored as env vars on **Render**.
9. Audit logging of significant actions.
10. No logging of raw sensitive data or credentials.

11. UI/UX Requirements (Per Screen)

Screens are listed with their purpose, key controls, and role visibility. Role symbols: A=Admin, SS=Store Staff, SP=Sales Personnel, SK=Senior Stakeholder.

11.1 Login

- Email + password **or email OTP (magic link)** sign-in (Supabase Auth); show role-aware redirect after login; error on invalid creds; lockout message if locked.

11.2 Dashboard (role-aware)

- A/SK: inventory health KPIs + sales summary + low-stock alerts (widgets per FR-DSH).
- SS: own-store stock status + quick actions (stock-in/out/transfer).
- SP: own sales period + store performance.
- Widgets clickable → drill-down grid.

11.3 Product Master (A)

- Grid: SKU, name, category, unit, sale price, cost price (masked to non-authorized roles), status. Search/filter/pagination. Create/edit modal. Archive button (confirm).

11.4 Inventory (Stock) Screen (A, SS own-store)

- Per-location stock grid: product, on-hand, safety, reorder, target, status badge, last-updated.
- Actions: Stock-In, Stock-Out, Transfer, Adjust (SS limited to own store).
- Each action = form modal: product, qty, (from/to) location, reason/notes; shows live validation and post-mutation feedback.

11.5 Sales Screen (A, SP)

- Record sale: store, date, line-item table (product autocomplete, qty, price), computed totals; validate & save; shows stock sufficiency errors inline.
- Sale history list with detail.

11.6 Sales Reports (A, SP, SK)

- Tabs: Daily / Quarterly / Yearly. Filters: store, product, date range/quarter/year. Table + charts (value & units). Export CSV/PDF.

11.7 Safety Stock / Reorder Config (A)

- Table per product-location: safety, reorder, target, auto-order toggle. Edit inline/modal. "Suggest by AI" button → shows AI recommendation modal (accept/modify/reject).

11.8 Supplier Master (A)

- Grid + create/edit; product mapping; supplier performance score column (value-add).

11.9 Purchase Orders (A; SS = receive)

- PO list: number, supplier, destination, status, expected date, progress. PO detail with line items and receive button (records goods-in). Create manual PO form.

11.10 Alerts Center (A, SS, SP role-scoped)

- Unread/read lists; filter by type; click-through to relevant record.

11.11 Executive Dashboard (SK, A)

- Consolidated: total stock value, turnover, low/out counts, day/qtr/yr sales, store comparison charts; drill-down to store/product.

11.12 Reports Export

- Consistent CSV/PDF export with column set per report; respects masking.

11.13 User Management (A)

- User table, create/edit, role & store assignment, activate/deactivate, password reset.

11.14 Audit Log Viewer (A)

- Filters (actor, action, date, entity); read-only display.

11.15 Onboarding Wizard (A)

- 6-step wizard (BRD §10.17) with progress + skip/resume.

11.16 AI Recommendation Center (A, SK)

- Pending & history of AI recommendations with rationale; accept/modify/reject actions.

12. Use Cases

UC-01 Record a Sale (Primary: SP)

Precondition: SP logged in. Main: choose store → add items → save → stock decreases (atomic `fn_mutate_stock`) → sales totals update (recompute triggers) → low-stock evaluation. Post: sale immutable, `SAL-<YYYY>-<seq>` number assigned; audit logged.

Alternate: insufficient stock → blocked with product detail (`409 INSUFFICIENT_STOCK`).

UC-01a Void / Return a Sale (A / SP)

Void: Admin enters reason → sale marked `voided` (excluded from reports), un-returned stock restored, audit `sale_voided`. Return: select original sale lines + qty → stock restored via `sale_return`, refund recomputed, returns cannot exceed sold qty and are blocked on voided sales.

UC-02 Stock-In / Receive Goods (SS, A)

Both ad-hoc stock-in and PO-linked receipt. Post: stock increases; PO status updates.

UC-03 Auto-Reorder at Reorder Point (System + A)

Trigger when $qty \leq \text{reorder}$. Auto-PO created per §8.1. Alerts: `LOW_STOCK` + pending PO to Admin.

UC-04 Generate AI Warehouse Recommendation (A/SK)

Select warehouse/product set → view rationale → accept/modify/reject → stored to audit.

UC-05 Cycle Count Adjustment (SS/A)

Enter counted qty → see variance → confirm → stock updated + audit reason.

UC-06 Transfer Between Locations (SS/A)

Choose from/to → validate → both rows updated atomically → audit.

UC-07 Run Daily/Quarterly/Yearly Report (SP/A/SK)

Select period & filters → view metrics (DB-first: daily via `daily_sales_summary` view; quarterly/yearly via aggregates) → export CSV/PDF.

UC-08 Manage User & Roles (A)

Create/edit user with role+store; deactivate; permission enforcement verified.

UC-09 Onboarding New Product (A)

Create SKU → set safety/reorder/target → map supplier + lead time + auto-order → initial stock → tracked.

UC-10 View Executive Dashboard (SK/A)

Consolidated KPIs → drill to store/product.

13. Acceptance & Verification (Testability)

Each BRD acceptance criterion (AC-1...AC-14) is verified by one or more **test cases** here (mapped in Section 14).

Test ID	Test Case	Expected Result
T-AC1	View stock for all products across 3 stores + warehouse	Grid shows correct qty & status, ≤3s
T-AC2	Perform Stock-In/Out/Transfer/Adjust	qty updates correctly; no negative stock; audit logged
T-AC3	Record sale → check stock & reports	Sale total correct; stock decremented; daily/qtr/yr reports reflect sale
T-AC4	Configure safety stock; set stock below reorder	LOW_STOCK alert created; reorder evaluation runs
T-AC5	Enable auto-order; set stock ≤ reorder	Auto-PO generated with correct qty (per §8.1)
T-AC6	Complete PO end-to-end	PO status transitions track; goods-in increases stock
T-AC7	Run AI warehouse recommendation	Rationale shown; accept/modify/reject works & logged
T-AC8	Sales personnel view store performance	Per-store + comparison correct
T-AC9	Senior stakeholder opens executive dashboard	KPIs + drill-down work, near real-time
T-AC10	Login & RBAC	Email/password and email OTP (magic link) sign-in work (Supabase Auth); users see only permitted data/menus; unauthorized API → 403
T-AC11	Check masking	Cost price masked to unauthorized roles & in DB/exports
T-AC12	Verify hosting	App live on Vercel + Render (Node backend/AI) + Supabase (postgres/auth/storage) free tier; DB working
T-AC13	Performance	Dashboards ≤3s p95; reports reflect within 5 min
T-AC14	Out-of-scope confirm	Client sign-off in BRD §9

14. Traceability Matrix

BRD ID	SRS Section	Test
BR-01 / FR-INV-03	§4.1.3	T-AC1
BR-02 / FR-INV-04–07	§4.1.4–7	T-AC2
BR-03 / FR-SAL-02	§4.2.2	T-AC3
BR-04 / FR-SST	§4.3	T-AC4
BR-05 / FR-AI-01	§8.1	T-AC5
BR-06 / FR-AI-02	§8.2	T-AC7
BR-07 / FR-INV-04...07	§4.1.4–7	T-AC2
BR-08 / FR-SAL-03	§4.2.3	T-AC8

BRD ID	SRS Section	Test
BR-09 / FR-MON-02	§11.11	T-AC9
BR-10 / Data	§6	T-AC12
BR-11 / FR-SEC	§7	T-AC11
BR-12 / Hosting	§2.3, §5	T-AC12
BR-13 / 3 stores	§4.1.2	T-AC1
BR-14 / FR-USER	§4.6	T-AC10
BR-15 / Alerts	§4.1.8	T-AC4
BR-16 / Audit	§4.5.8	T-AC2
Value-add modules	§4.5	Partial suite

15. Open / TBD Items

#	Item	Decision Needed By	Owner
TBD-1	Backend: Node.js vs Python — DECIDED: Node.js (BRD v3.4)	Resolved	SA (Anoop) + Tech Lead
TBD-2	Component & charting libraries — DECIDED: shadcn/ui + Recharts (BRD v3.4)	Resolved	SA/Tech Lead
TBD-3	Exact masked-field list confirmed by client	SRS sign-off	Client
TBD-4	AI auto-qty mode default (target-based vs forecast-based)	Sprint 3	SA
TBD-5	Perishable (FR-EXP) inclusion in v1.0 (Could)	Sprint planning	SM/PM
TBD-6	Alert email provider/limit on free tier	Sprint 3	Tech Lead (may use Supabase Auth email / provider SMTP)
TBD-7	AI auto-ordering + warehouse recommendations (Groq) — planned Sprint 3 ; schema + alert plumbing already live (<code>alerts</code> , <code>safety_stock_rules</code> , <code>ai_recommendations</code> , <code>app_settings</code> thresholds)	Sprint 3	SA/Tech Lead

Approval Sign-off

Role	Name	Signature	Date
Client / Sponsor	Prof		
Project Manager	Laxman Patel		
Business Analyst	Ved Naik		
Solutions Architect	Anoop Gupta		

Role	Name	Signature	Date
Scrum Master	Devdarshan S		
Tech Lead	Rohan Vashisht		

End of SRS — Version 1.2 · Project: Vaultory · Team: Vaultory